The Division of Research
Executed-Protocol Review Companion

6 October 2026

Status. This author-review companion supplies the detailed reproducibility material and cached-book assignment analysis accompanying the revised manuscript. It is separate from the manuscript PDF and is not an assertion that an additional conference submission or appendix is permitted. No new LLM generations were collected for these revisions.

1 Audited reproducibility specification

This specification applies to the final audit-corrected primary experiment, not the superseded pre-audit collections. The input panel comprises 34 month-end labels from July 2023 through April 2026. The 12 development and 22 complementary dates share the corrected SEC-history packet schema. No model calls were made to produce this record.

1.1 Universe, routes, and access matrices

The fixed investable universe, in request order, is AAPL, MSFT, NVDA, GOOGL, META, AMZN, TSLA, WMT, PG, JPM, BAC, XOM, JNJ, UNH. Ticker, company, and sector are supplied to every seat, outside the 32 partitioned research fields. The common-core content is:

Ticker	Company	Sector
AAPL	Apple	Information Technology
MSFT	Microsoft	Information Technology
NVDA	NVIDIA	Information Technology
GOOGL	Alphabet	Communication Services
META	Meta Platforms	Communication Services
AMZN	Amazon	Consumer Discretionary
TSLA	Tesla	Consumer Discretionary
WMT	Walmart	Consumer Staples
PG	Procter & Gamble	Consumer Staples
JPM	JPMorgan Chase	Financials
BAC	Bank of America	Financials
XOM	Exxon Mobil	Energy
JNJ	Johnson & Johnson	Health Care
UNH	UnitedHealth Group	Health Care

The executed common-core text uses left-aligned character widths 7, 22, and 26, respectively, with a header row and one newline between rows.

Table 1 separates the requested endpoint from its stored catalog canonical slug and the observed serving provider. These are route identities, not a guarantee of immutable closed weights or bitwise-reproducible future service responses. The catalog was retrieved on 21 July 2026 at 01:07:39 UTC. Provider sets below cover the 238 shared, 3,332 audited $K = 7$, and 9,520 audited $K = 2, \ldots, 6$ primary cached rows; they do not pool legacy or robustness-only collections.

Table 1: Frozen requested model routes, catalog identities, and observed serving providers. Response-model strings equal the requested strings in these primary caches.

Route label	Requested endpoint	Catalog canonical slug	Observed providers
openai	`openai/gpt-4o`	`openai/gpt-4o`	OpenAI
google	`google/gemini-2.5-flash`	`google/gemini-2.5-flash`	Google
meta	`meta-llama/llama-3.1-70b-instruct`	`meta-llama/llama-3.1-70b-instruct`	DeepInfra, WandB
alibaba	`qwen/qwen3-235b-a22b-2507`	`qwen/qwen3-235b-a22b-07-25`	AtlasCloud, DeepInfra, Friendli, Google, Nebius, Novita, Parasail
mistral	`mistralai/mistral-large`	`mistralai/mistral-large`	Mistral
deepseek	`deepseek/deepseek-chat`	`deepseek/deepseek-chat-v3`	DeepInfra
anthropic	`anthropic/claude-sonnet-4.5`	`anthropic/claude-4.5-sonnet-20250929`	Amazon Bedrock, Anthropic

Tables 2–5 enumerate each field once in the frozen full-packet order. Their C and M columns give the coherent and mixed seat numbers, respectively, and thus specify both complete 7×32 binary access matrices without omitting zero entries:

$$A^L_{si} = \mathbf{1}\{s = L_i\}, \qquad L \in \{\mathrm{C}, \mathrm{M}\}, \quad s = 1, \ldots, 7, \quad i = 1, \ldots, 32.$$

Both layouts have row sums $(4, 2, 3, 5, 7, 7, 4)$ and column sums one. The shared layout has every entry one. Seat labels denote information positions, not permanently attached model routes: the $K = 7$ analysis averages all 7! bijections between routes and seats.

For exact within-packet column order, the frozen coherent seat lists, expressed using these field indices, are

1: 1, 2, 3, 5

2: 4, 6

3: 7, 8, 9

4: 10, 11, 12, 13, 14

5: 15, 16, 17, 18, 19, 20, 21

6: 22, 23, 24, 27, 28, 29, 31

7: 25, 26, 30, 32

The mixed seat lists are

1: 3, 16, 32, 9

2: 19, 10

3: 11, 30, 18

4: 22, 14, 28, 2, 17

5: 1, 24, 13, 15, 7, 4, 27

6: 25, 20, 8, 6, 12, 31, 26

7: 29, 23, 5, 21

The frozen assignment SHA-256 is `62dac5bf0f016fef5c40159f02ad5b097228c80f801b140f938584d6fea35cf3`.

1.2 Field definitions and availability

Definitions below distinguish stored numerical transformations from prompt rendering. In the tables, P means multiply the raw fraction by 100 and print two decimal places; N means print two decimal places with no scaling; I means `int(round(float(value)))`; T means trim the text, substituting `none` for an empty string. All formats first render `None` or a non-finite floating value as `NA`. No cross-sectional standardization, ranking, winsorization, or numeric imputation is applied by the packet renderer.

Prices and windows. P_t denotes the last available repository adjusted close at or before the snapshot label. Market features first select the 14 investable columns, forward-fill prices, and discard remaining incomplete rows; at least 253 complete rows are required. Session counts are observation counts, not calendar days. The labels `ret_7d`, `ret_30d`, and `ret_90d` correspond to 5, 21, and 63 sessions. The valuation and dollar-volume series instead uses Yahoo quote close, dividend-unadjusted and split-adjusted to the recorded corporate-action basis. Volume missing on a selected date is zero-filled before computing dollar volume.

Fundamentals. All eligible SEC facts have both filing date and period end no later than the snapshot label, finite values, and form 10-Q, 10-K, 10-Q/A, or 10-K/A. For an instant fact, select the maximum tuple (filing date, period end, negative tag-priority index, accession number). For annual facts, retain 10-K/10-K/A observations with a start date and a 300–390-calendar-day period; select the maximum (filing date, negative tag-priority index, accession number) within each period end, then sort by period end. "Latest annual" and "previous annual" mean the final two such distinct period ends, not trailing-twelve-month reconstructions. The exact ordered tag candidates are given in Table 6.

Write A, E, C, D, R, N for the selected assets, equity, cash, debt, annual revenue, and annual net income. Shares use the selected instant shares fact, falling back only when absent to the latest eligible annual weighted-average shares fact. Multiply as-filed shares by the product of split factors strictly after the shares' measurement date, through the recorded corporate-action cutoff (20 July 2026). Market capitalization M is this share count times the quote close on the same split basis. Thus the availability filter concerns historical filings; the price/share basis is harmonized using a later recorded corporate-action cutoff, not claimed to be a historical download-vintage archive. Ratios return NA if either operand is missing, the absolute denominator is at most 10^{-12}, or the result is non-finite. Except for P/E's positive-income requirement, negative denominators are not suppressed. No maximum staleness filter is applied to eligible facts.

Events. The corrected SEC event history joins the recent table with all named historical submission tables and deduplicates on accession number, filing date, form, and primary document. Only filing dates at or before the snapshot label enter. All event-window ages are inclusive calendar-day ages. Counts can legitimately be zero; absent recency values are NA. For a same-day latest-event tie, the sorted history's first matching row is retained. Known item labels are: 1.01, material agreement; 1.02, agreement termination; 2.01, acquisition or disposal; 2.02, earnings release; 2.05, restructuring costs; 2.06, material impairment; 5.02, director or officer change; 7.01, regulation FD disclosure; and 8.01, other material event. Unknown codes retain the literal prefix `item` followed by the code.

Risk and macro series. Risk estimation uses the last 253 rows of the forward-filled adjusted-close file and daily simple returns, dropping incomplete return rows. Its equal-weight reference averages *all 50 columns in that file*, not only the 14 investable stocks and not SPY. Those columns are AAPL, MSFT, NVDA, AMZN, GOOGL, META, BRK-B, LLY, AVGO, JPM, TSLA, V, XOM, UNH, MA, COST, HD, PG, NFLX, JNJ, ABBV, BAC, KO, CRM, WMT, ORCL, CSCO, CVX, MRK, AMD, PEP, TMO, LIN, MCD, DIS, ABT, QCOM, IBM, ADBE, GE, CAT, NOW, AXP, TXN, VZ, PM, INTU, AMGN, ISRG, SPGI. Variance, covariance, and standard deviation use sample normalization $(n-1)$. A beta denominator at most 10^{-12} (or a non-finite denominator failing that comparison) yields zero; other non-finite results render as NA. Macro observations are finite Yahoo quote closes, restricted to dates no later than the snapshot label; levels take the last such observation and returns require at least 64 observations. There is no staleness cap. The SPY and HYG macro returns use quote closes, not adjusted-close total returns. Yield levels are used as supplied, with no extra factor of ten or 100. The curve spread is formed from unrounded levels and only then rounded for display.

Table 2: Market fields, transformations, and access-matrix columns. C/M are coherent/mixed seats; format codes are defined in the text.

i	Field ID / displayed header	C	M	Fmt	Definition (raw units)
1	`market.ret_7d` `ret_7d%`	1	5	P	Adjusted-close simple return over 5 sessions: P[t]/P[t-5]-1; the historical label is ret_7d. (fraction).
2	`market.ret_30d` `ret_30d%`	1	4	P	Adjusted-close simple return over 21 sessions: P[t]/P[t-21]-1. (fraction).
3	`market.ret_90d` `ret_90d%`	1	1	P	Adjusted-close simple return over 63 sessions: P[t]/P[t-63]-1. (fraction).
4	`market.vol_30d` `vol_30d%`	2	5	P	Sample standard deviation of the last 21 daily log returns, multiplied by sqrt(252). (annualized fraction).
5	`market.drawdown_90d` `dd_90d%`	1	7	P	Current adjusted close divided by the maximum of the last 63 closes, minus one; not the worst peak-to-trough loss within that window. (fraction).
6	`market.adv_bn` `ADV_$bn`	2	6	N	Mean of the last 21 products of split-adjusted, dividend-unadjusted quote close and volume, divided by 1e9; missing volume is zero. (USD billions/day).

Table 3: Fundamentals fields, transformations, and access-matrix columns. C/M are coherent/mixed seats; format codes are defined in the text.

i	Field ID / displayed header	C	M	Fmt	Definition (raw units)
7	`fundamentals.pe` `P/E`	3	5	N	Split-consistent market capitalization M divided by latest eligible annual net income N; NA if N is missing or nonpositive. (multiple).
8	`fundamentals.price_to_sales` `P/S`	3	6	N	M divided by latest eligible annual revenue R. (multiple).
9	`fundamentals.price_to_book` `P/B`	3	1	N	M divided by selected stockholders equity E. (multiple).
10	`fundamentals.roa` `ROA%`	4	2	P	Latest eligible annual net income N divided by selected assets A; not trailing-quarter or TTM income. (fraction).
11	`fundamentals.debt_to_assets` `debt/A%`	4	3	P	Selected long-term debt/finance-lease fact D divided by A; not a constructed total-debt measure. (fraction).
12	`fundamentals.cash_to_assets` `cash/A%`	4	6	P	Selected cash fact C divided by A. (fraction).
13	`fundamentals.revenue_growth` `rev_g%`	4	5	P	(Latest eligible annual revenue R minus previous annual revenue R_prev) divided by abs(R_prev). (fraction).
14	`fundamentals.accounting_age` `age_d`	4	4	I	Calendar days from the filing date of the selected latest annual revenue fact to the snapshot label; not days since the accounting period ended. (calendar days).

Table 4: Events fields, transformations, and access-matrix columns. C/M are coherent/mixed seats; format codes are defined in the text.

i	Field ID / displayed header	C	M	Fmt	Definition (raw units)
15	`events.eight_k_30d` `8K_30d`	5	5	I	Count of 8-K and 8-K/A filings with filing age between 0 and 30 calendar days, inclusive. (count).
16	`events.eight_k_90d` `8K_90d`	5	1	I	Count of 8-K and 8-K/A filings with filing age between 0 and 90 calendar days, inclusive. (count).
17	`events.earnings_release_90d` `earn_90d`	5	4	I	Count of item-code 2.02 occurrences across the eligible 90-day filings; counts item occurrences, not distinct event days. (count).
18	`events.material_event_90d` `material`	5	3	I	Count of item-code occurrences in {1.01,1.02,2.01,2.05,2.06,5.02,8.01} across eligible 90-day filings. (count).
19	`events.days_since_latest_8k` `last_8K_d`	5	2	I	Calendar days since latest 8-K/8-K/A within the prior 90 days; NA if none exists in that window. (calendar days).
20	`events.days_since_periodic_filing` `periodic_d`	5	6	I	Calendar days since latest 10-Q or 10-K filed by the snapshot label; no lookback cap and no amended forms in this field. (calendar days).
21	`events.latest_8k_items` `latest_event`	5	7	T	First three mapped item labels, in source order, of latest eligible 90-day 8-K/8-K/A, joined by semicolon-space; unknown code becomes item <code>; none in prior 90d if no labels. (text).

Table 5: Risk fields, transformations, and access-matrix columns. C/M are coherent/mixed seats; format codes are defined in the text.

i	Field ID / displayed header	C	M	Fmt	Definition (raw units)
22	`risk.market_beta` `beta`	6	4	N	Sample covariance of stock and equal-weight 50-stock reference daily simple returns divided by reference sample variance, over at most 252 returns. Zero if variance <= 1e-12. (dimensionless).
23	`risk.downside_beta` `down_beta`	6	7	N	Same beta calculation restricted to days with negative reference return; denominator is the sample variance on that subset. Zero if variance <= 1e-12. (dimensionless).
24	`risk.market_corr` `mkt_corr`	6	5	N	Pearson correlation between stock and reference daily simple returns over the risk window. (dimensionless).
25	`risk.idio_vol` `idio_vol%`	7	6	P	Sample standard deviation of stock return minus estimated beta times reference return, multiplied by sqrt(252); no intercept fitted. (annualized fraction).
26	`risk.max_drawdown_1y` `max_dd%`	7	6	P	Minimum of P[u]/max_{v<=u} P[v]-1 over the last 253 available closes; running peak starts at the window beginning. (fraction).
27	`risk.ten_year_yield` `10y%`	6	5	N	Last finite Yahoo ^TNX quote close at or before the snapshot label, used directly without an extra scale factor. (percent yield level).
28	`risk.three_month_yield` `3m%`	6	4	N	Last finite Yahoo ^IRX quote close at or before the snapshot label, used directly without an extra scale factor. (percent yield level).

29	`risk.yield_curve_`
	`spread`
	`10y-3m_pp`
		6	7	N	Ten-year yield minus three-month yield, derived before display rounding; NA if either source value is missing. (percentage points).

30	`risk.vix` `VIX`	7	3	N	Last finite Yahoo ^VIX quote close at or before the snapshot label. (index points).

31	`risk.equity_market_`
	`return_3m`
	`SPY_3m%`
		6	6	P	Simple return over 63 available SPY quote-close observations: Q[t]/Q[t-63]-1; not adjusted-close total return. NA with fewer than 64 observations. (fraction).

32	`risk.high_yield_`
	`return_3m`
	`HYG_3m%`
		7	1	P	Simple return over 63 available HYG quote-close observations: Q[t]/Q[t-63]-1; not adjusted-close total return. NA with fewer than 64 observations. (fraction).

Table 6: Ordered SEC tag candidates. Instant shares consider both `us-gaap` and `dei`; all other lookups use `us-gaap`. Source selection follows the date/priority ordering in the text, rather than treating a lower-priority tag as usable only when a higher-priority tag is wholly absent.

Quantity	Unit	Ordered tags
A	USD	`Assets`
E	USD	`StockholdersEquity`; `StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest`
C	USD	`CashAndCashEquivalentsAtCarryingValue`; `CashCashEquivalentsRestrictedCashAndRestrictedCashEquivalents`
D	USD	`LongTermDebtAndFinanceLeaseObligations`; `LongTermDebt`; `LongTermDebtNoncurrent`
Instant shares	shares	`CommonStockSharesOutstanding`; `EntityCommonStockSharesOutstanding`
Annual shares fallback	shares	`WeightedAverageNumberOfDilutedSharesOutstanding`; `WeightedAverageNumberOfSharesOutstandingBasic`
R	USD	`RevenueFromContractWithCustomerExcludingAssessedTax`; `Revenues`; `SalesRevenueNet`
N	USD	`NetIncomeLoss`; `ProfitLoss`

1.3 Executed prompt and output contract

The following is the entire executed user-message template, with the three source substitutions shown in braces. Line wrapping here is purely typographic. `snapshot['common_core']` expands to the fixed-width universe table above; `packet_text` expands as specified below; `example` is the first ticker, AAPL. Blank lines separate the source paragraphs. The historical final example has an extra closing brace (`]}}.`); it is reproduced rather than repaired.

```
You are an institutional U.S. equity analyst.  Construct a one-month active long/short
book that seeks the best expected risk-adjusted return from the supplied evidence.  Do
not create different positions merely to appear diversified.

You receive one packet from a research workflow.  Its field composition was fixed before
model collection.  Use only the supplied packet.  Do not infer missing fields, import
current facts, or use outside knowledge.  The calendar date is intentionally hidden.

COMMON UNIVERSE
{snapshot['common_core']}

PERMITTED INFORMATION
{packet_text}

Give every listed ticker a signed conviction score from -1 to +1, with both positive and
negative scores.  Positive means long and negative means short.  These are research
scores, not final capital weights.  The fixed evaluation pipeline subtracts their mean
and rescales absolute exposure to 1, imposing the same dollar-neutral book construction
on every response.

Reply with one JSON object and no markdown.  The weights field must be an object
containing every listed ticker exactly once.  Use this schema:
{"weights":{"{example}":0.10,"...":0.00},"thesis":"at most 35
words","primary_signals":["signal 1","signal 2"]}}.
```

The packet starts with the literal line `RESEARCH PACKET`. If it has asset-specific fields, the next line is `ticker |` followed by their displayed headers joined with ` | `. Each of the 14 ticker rows follows in universe order, with formatted values joined identically. Macro fields are removed from those columns and appended once as a final line beginning `market context |` , followed by `header=value` pairs joined with ` | `. Both the asset and macro subsequences preserve their order within the frozen seat list. The full shared packet uses field-index order. Lines are joined by a single newline. There is no seat label, date, persona, architecture name, or omitted-field placeholder in this renderer.

The actual API `response_format` object is given below; this, not the prompt's illustrative ellipsis, defines all required output keys. The schema contains no numerical score bounds, thesis word-count limit, or fixed number of signals. Those are distinct from the prompt request. The collector separately rejects missing/unknown/duplicate symbols, non-finite values, absolute scores above $1+10^{-12}$, and a centered book with absolute sum at most 10^{-12}. It does not separately enforce a positive/negative-sign mix or the 35-word thesis request.

```
{
  "type": "json_schema",
  "json_schema": {
    "name": "active_book",
    "strict": true,
    "schema": {
      "type": "object",
      "properties": {
        "weights": {
          "type": "object",
          "properties": {
            "AAPL": {
              "type": "number"
            },
            "MSFT": {
              "type": "number"
```

```
          },
          "NVDA": {
            "type": "number"
          },
          "GOOGL": {
            "type": "number"
          },
          "META": {
            "type": "number"
          },
          "AMZN": {
            "type": "number"
          },
          "TSLA": {
            "type": "number"
          },
          "WMT": {
            "type": "number"
          },
          "PG": {
            "type": "number"
          },
          "JPM": {
            "type": "number"
          },
          "BAC": {
            "type": "number"
          },
          "XOM": {
            "type": "number"
          },
          "JNJ": {
            "type": "number"
          },
          "UNH": {
            "type": "number"
          }
        },
        "required": [
          "AAPL",
          "MSFT",
          "NVDA",
          "GOOGL",
          "META",
          "AMZN",
          "TSLA",
          "WMT",
          "PG",
          "JPM",
          "BAC",
          "XOM",
          "JNJ",
          "UNH"
        ],
        "additionalProperties": false
      },
      "thesis": {
        "type": "string"
      },
      "primary_signals": {
        "type": "array",
        "items": {
          "type": "string"
        }
      }
    },
    "required": [
      "weights",
      "thesis",
      "primary_signals"
    ],
    "additionalProperties": false
  }
 }
}
```

The request is sent to `https://openrouter.ai/api/v1/chat/completions`, with the requested model string, temperature 0.7, `max_tokens=1400`, one `user` message and no separate system message, `provider={"require_parameters":true}`, and the schema above. A seed is supplied

only when the catalog advertises it (all routes here except Anthropic). For the pipe-joined identity `date|architecture|team_size|packet_index|vendor`, the base seed is 2026072100 plus the integer represented by the first seven hexadecimal SHA-256 characters, modulo 100,000,000. Shared uses architecture `shared` and packet index zero; the audited partitions use `mixed_audited` or `coherent_audited`, with one-based packet indices. For five Google curve requests an additional recovery offset of one is recorded: (2023-10-31, mixed, $K = 5$, seat 2), (2024-03-31, mixed, $K = 5$, seat 2), (2024-08-31, mixed, $K = 5$, seat 3), (2024-09-30, coherent, $K = 3$, seat 2), and (2024-11-30, coherent, $K = 3$, seat 2). The collector permits at most five attempts per job; ordinarily the same payload is retried, and parse/format diagnostics and response metadata are retained. These controls do not guarantee provider-side seed determinism. System fingerprints are absent in 10,047 of the 13,090 primary rows, although provider and response-model strings are present in all those rows.

Supplied machine-readable objects. The accompanying `repro_protocol.json` supplies both literal 7×32 matrices, the ordered seat lists, complete route/provider metadata, field definitions, rendering rules, universe, and seed offsets. `repro_response_format.json` contains the API schema; `repro_prompt_template.txt` preserves the template without PDF line wrapping. `repro_prompt_examples.json` supplies the complete, unabridged shared, mixed-seat-1, and coherent-seat-1 rendered prompts for 31 July 2023, together with the recorded source, packet, prompt, and request-payload hashes. These are supplied objects, not promises of a future supplement.

2 Within-date assignment dispersion

For each of 34 dates and each split design, all 7! = 5,040 bijections between the seven requested routes and the seven frozen packets are enumerated using archived books. Shared access supplies one same-date seven-route reference. The assignment varies the route-to-packet mapping, not the partition of fields. No assignment is treated as an independent observation for inference.

Within each date, we compute the empirical 5th, 50th, and 95th percentiles of effective-book breadth and of its paired gap against shared access. Table 7 averages each date-level percentile across dates; it does not report pooled percentiles or confidence intervals. Mean breadth was checked against both frozen audited result objects, with maximum absolute discrepancy below 10^{-15}.

Table 7: Descriptive dispersion over route-to-packet assignments.

Assignment sensitivity	Mixed	Coherent
N_{eff}: $q_{.05}/q_{.50}/q_{.95}$	3.35 / 3.97 / 4.62	3.20 / 3.89 / 4.58
ΔN_{eff}: $q_{.05}/q_{.50}/q_{.95}$	+1.30 / +1.92 / +2.57	+1.15 / +1.84 / +2.53
Above shared: mean date fraction	99.666%	98.243%
Above shared: lowest date fraction	90.24%	73.13%
Smallest single-assignment ΔN_{eff}	-0.838	-1.079
Date of smallest gap	2024-06-30	2024-08-31
Dates with any negative gap	5/34	7/34
Assignments below shared	573	3,010

All 7! = 5,040 one-to-one route-to-packet assignments per design and date; 34 dates. The first two rows are arithmetic means across dates of within-date quantiles, not pooled quantiles or confidence intervals. Δ subtracts that date's shared N_{eff}. Fractions use strict $>$; counts below shared are out of 171,360 configurations per design, not independent observations. No new inference is performed.

The assignment-averaged mixed and coherent gaps are positive on every date, but individual assignments need not beat shared access. Mixed has 573 below-shared assignments over 5 dates; coherent has 3010 over 7 dates, out of 171,360 assignments per design. There are no ties. The minimum paired gaps are -0.838 on 2024-06-30 and -1.079 on 2024-08-31. The lowest date-level fractions above shared are 90.24% and 73.13%, respectively. These descriptive tails qualify, rather than contradict, the positive assignment-averaged result.

The full per-date distributions, minimizing assignments, counts, source hashes, and frozen-mean checks are supplied in `information_capacity_assignment_dispersion_results.json`. The repository command `python3 src/cohort_information_capacity_assignment_dispersion.py` reproduces the analysis from the existing caches without contacting a model service.